

A central angle of a circle subtends an arc length of 5π cm.

Given the circle has a radius of 9 cm, find the measure of the central angle in degrees.

Solution

$$s = \theta r$$

$$5\pi = \theta(9)$$

$\frac{1}{2}$ mark for substitution into correct formula

$$\theta = \frac{5\pi}{9}$$

$\frac{1}{2}$ mark for solving for θ

$$\theta \text{ (in degrees)} = \frac{5\pi}{9} \cdot \frac{180^\circ}{\pi}$$

$$= 100^\circ$$

1 mark for conversion to degrees

2 marks

Solve the equation $\csc^2 \theta + 3 \csc \theta - 4 = 0$ over the interval $[0, 2\pi]$.
Express your answers as exact values or correct to 3 decimal places.

Solution**Method 1**

$$\csc^2 \theta + 3 \csc \theta - 4 = 0$$

$$(\csc \theta - 1)(\csc \theta + 4) = 0$$

$$\csc \theta = 1$$

$$\csc \theta = -4$$

1 mark for solving for $\csc \theta$

$$\sin \theta = 1$$

$$\sin \theta = -\frac{1}{4}$$

1 mark for reciprocal of $\csc \theta$

$$\theta = \frac{\pi}{2}$$

$$\theta_r = 0.252\ 680$$

or

$$\theta = 1.570\ 796$$

$$\theta = 3.394\ 273, 6.030\ 505$$

$$\theta = \frac{\pi}{2}, 3.394, 6.031$$

2 marks (1 mark for consistent solutions of each trigonometric equation)

or

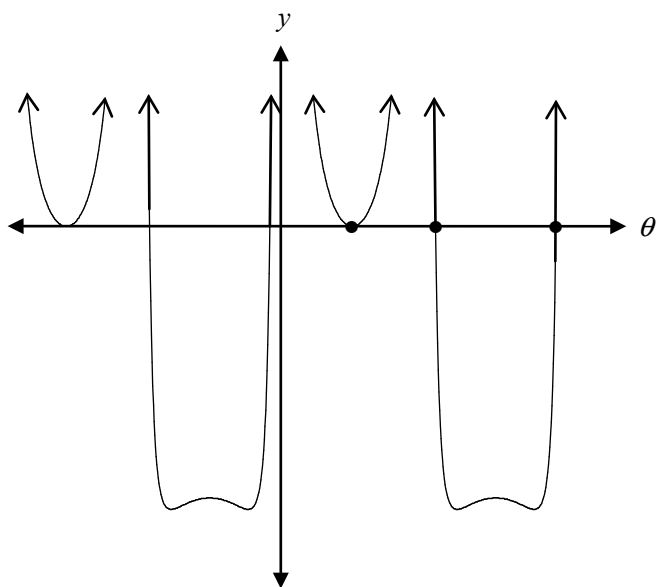
$$\theta = 1.571, 3.394, 6.031$$

4 marks

Method 2–Graphing Calculator

$$y = \left(\frac{1}{\sin \theta} \right)^2 + \frac{3}{\sin \theta} - 4$$

1 mark for equation



1 mark for justification

Find all zeros from $[0, 2\pi]$.

1 mark for restricted domain

$$\theta = 1.571, 3.394, 6.031$$

1 mark for solutions

4 marks

Jess invests \$12 000 at a rate of 4.75% compounded monthly.

How long will it take for Jess to triple her investment?

Express your answer in years, correct to 3 decimal places.

Solution

Method 1

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$36\,000 = 12\,000\left(1 + \frac{0.0475}{12}\right)^{12t} \quad \frac{1}{2} \text{ mark for substitution}$$

$$3 = \left(1 + \frac{0.0475}{12}\right)^{12t}$$

$$\ln 3 = \ln\left(1 + \frac{0.0475}{12}\right)^{12t} \quad \frac{1}{2} \text{ mark for applying logarithms}$$

$$\ln 3 = 12t \ln\left(1 + \frac{0.0475}{12}\right) \quad 1 \text{ mark for power rule}$$

$$t = \frac{\ln 3}{12 \ln\left(1 + \frac{0.0475}{12}\right)} \quad \frac{1}{2} \text{ mark for isolating } t$$

$$t = 23.174\,425$$

$$t = 23.174 \text{ years} \quad \frac{1}{2} \text{ mark for evaluating quotient of logarithms}$$

3 marks

Note(s):

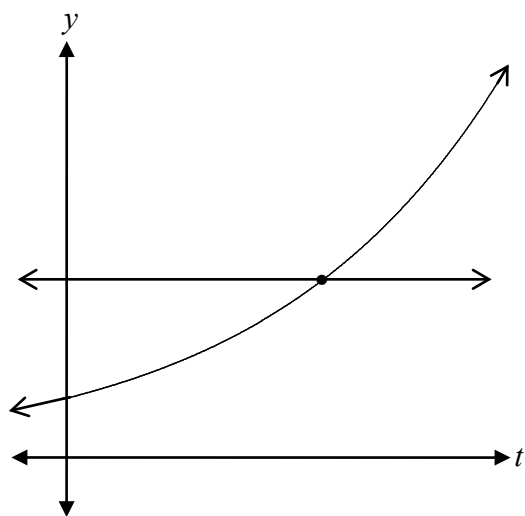
- award a maximum of 2 marks for the formula $A = Pe^{rt}$ used correctly

Method 2–Graphing Calculator

$$y = 3$$

$$y = \left(1 + \frac{0.0475}{12}\right)^{12t}$$

1 mark for equations

**or**

1 mark for justification

Find the value of t at the point of intersection of these two functions.

$$t = 23.174 \text{ years}$$

1 mark for solution

3 marks

The 4th term in the binomial expansion of $\left(qx^2 - \frac{3}{x}\right)^{10}$ is $414\,720x^{11}$.

Determine the value of q algebraically.

Solution

$$t_4 = {}_{10}C_3 \left(qx^2\right)^7 \left(-\frac{3}{x}\right)^3 \quad 2 \text{ marks (1 mark for } {}_{10}C_3, \frac{1}{2} \text{ mark for each consistent factor)}$$

$$414\,720x^{11} = 120 \left(q^7 x^{14}\right) \left(-\frac{27}{x^3}\right) \quad \frac{1}{2} \text{ mark for comparing coefficients}$$

$$414\,720 = -3240q^7$$

$$q^7 = -128$$

$$q = -2$$

$\frac{1}{2}$ mark for solving for q

3 marks

Question 5

P1

Bella has 2 pairs of shoes, 3 pairs of pants, and 10 shirts.

Carey has 4 pairs of shoes, 4 pairs of pants, and 4 shirts.

An outfit is made up of one pair of shoes, one pair of pants, and one shirt.

Who can make more outfits? Justify your answer.

Solution

Bella: $2 \times 3 \times 10 = 60$ outfits

Carey: $4 \times 4 \times 4 = 64$ outfits

$\therefore$ Carey can make more outfits.

1 mark for justification

1 mark

Question 6

P4

In the binomial expansion of $(x - y)^{10}$, how many terms will be positive?

Justify your answer.

Solution

Six terms will be positive.

1 mark for six terms

The term will be positive when “ $-y$ ” has an even exponent.

1 mark for justification

2 marks

Solve the following equation algebraically where $180^\circ \leq \theta \leq 360^\circ$.

$$2\sin^2 \theta + 5\cos \theta + 1 = 0$$

Solution

$$2(1 - \cos^2 \theta) + 5\cos \theta + 1 = 0$$

1 mark for identity

$$2 - 2\cos^2 \theta + 5\cos \theta + 1 = 0$$

$$2\cos^2 \theta - 5\cos \theta - 3 = 0$$

$$(2\cos \theta + 1)(\cos \theta - 3) = 0$$

$$\cos \theta = -\frac{1}{2}$$

$$\cos \theta = 3$$

1 mark for solving for $\cos \theta$

$$\theta_r = 60^\circ$$

$\therefore$ no solution

1 mark for indicating no solution

$$\theta = 240^\circ$$

1 mark for solving for θ

4 marks

Note(s):

- award a maximum of 3 marks if not solved algebraically

Solve the following equation algebraically:

$$\log_3(x-4) + \log_3(x-2) = 1$$

Solution

Method 1

$$\log_3(x-4) + \log_3(x-2) = 1$$

$$\log_3(x-4)(x-2) = 1$$

1 mark for product rule

$$3^1 = (x-4)(x-2)$$

1 mark for exponential form

$$3 = x^2 - 6x + 8$$

$$0 = x^2 - 6x + 5$$

$$0 = (x-5)(x-1)$$

$$x = 5 \quad \cancel{x=1}$$

½ mark for solving for x within a quadratic equation
½ mark for rejecting extraneous root

3 marks

Method 2

$$\log_3(x-4) + \log_3(x-2) = 1$$

$$\log_3(x-4)(x-2) = 1$$

1 mark for product rule

$$\log_3(x^2 - 6x + 8) = \log_3 3$$

½ mark for logarithmic form

$$x^2 - 6x + 8 = 3$$

½ mark for equating arguments

$$x^2 - 6x + 5 = 0$$

$$(x-1)(x-5) = 0$$

$$\cancel{x=1} \quad x = 5$$

½ mark for solving for x within a quadratic equation
½ mark for rejecting extraneous roots

3 marks

Given that $f(x) = \{(1, 3), (2, 5), (3, 4), (4, 2)\}$, find $f(f(3))$.

Solution

$$\begin{aligned} f(f(3)) &= f(4) \\ &= 2 \end{aligned}$$

½ mark for $f(3) = 4$

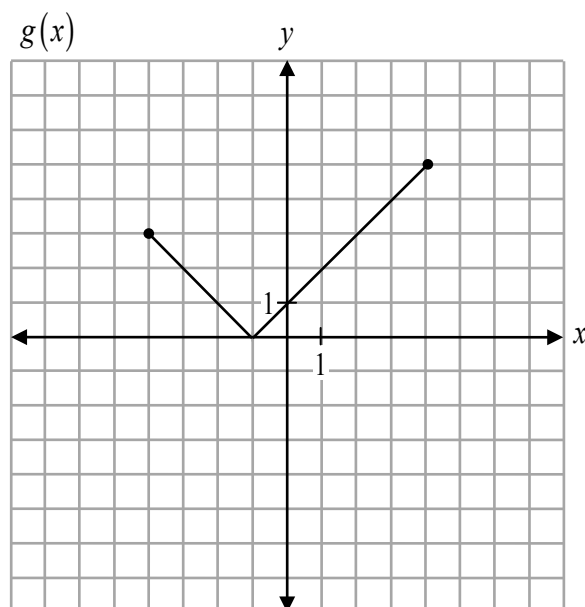
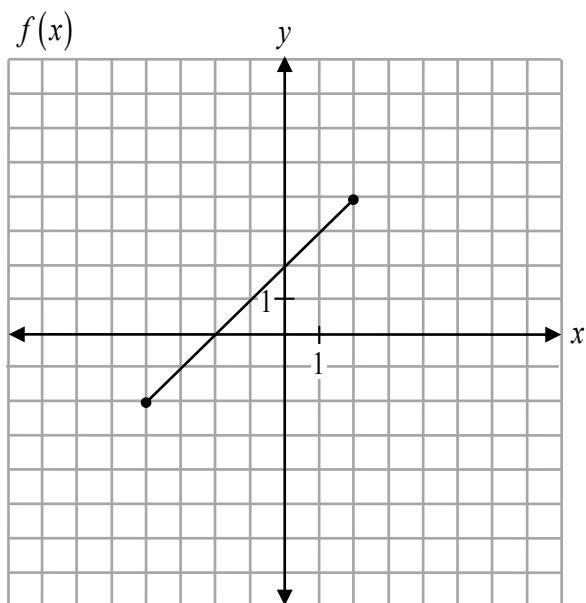
½ mark for $f(4) = 2$

1 mark

Question 10

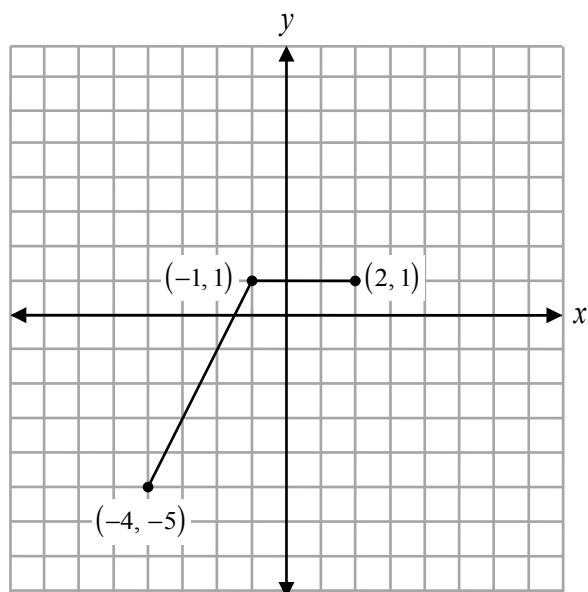
R1

Given the graphs of $f(x)$ and $g(x)$ below,



sketch the graph of $y = f(x) - g(x)$.

Solution



x	$f(x)$	$g(x)$	$f(x) - g(x)$
-4	-2	3	-5
-2	0	1	-1
-1	1	0	1
0	2	1	1
2	4	3	1

1 mark for subtraction of $f(x) - g(x)$

1 mark for restricted domain

2 marks

Question 11

R2, R3

Given the graph of $y = f(x)$, describe the transformations to obtain the graph of the function $y = f(2x - 6)$.

Solution

Method 1

Factor out the 2.

$$y = f(2(x - 3))$$

Horizontally compress by a factor of 2.
Then shift 3 units to the right.

1 mark for starting with a horizontal compression by a factor of 2
1 mark for ending with a horizontal shift of 3 units to the right

Method 2

$$y = f(2x - 6)$$

Shift 6 units to the right.
Then horizontally compress by a factor of 2.

1 mark for starting with a horizontal shift of 6 units to the right
1 mark for ending with a horizontal compression by a factor of 2

2 marks

2 marks

Question 12

R5

Given $f(x) = \{(-3, 4), (2, 7), (8, 6)\}$, state the domain of the resulting function after $f(x)$ is reflected through the line $y = x$.

Solution

Domain: $\{4, 6, 7\}$

1 mark for correct domain

1 mark

Note(s):

- award $\frac{1}{2}$ mark for stating the inverse of the function: $f^{-1}(x) = \{(4, -3), (7, 2), (6, 8)\}$

Question 13

R8

Determine the value of y in the following equation:

$$\log_x 27 - \log_x 3 = 2 \log_x y$$

Solution

$$\log_x 27 - \log_x 3 = 2 \log_x y$$

$$\log_x \frac{27}{3} = 2 \log_x y$$

1 mark for quotient rule

$$\log_x 9 = \log_x y^2$$

1 mark for power rule

$$9 = y^2$$

$$y = \pm 3$$

$$y = 3 \quad \cancel{y = -3}$$

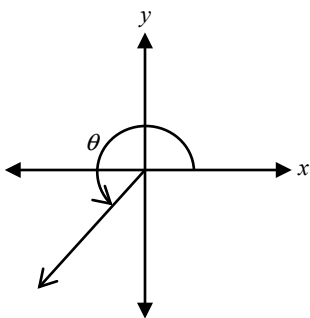
 $\frac{1}{2}$ mark for positive value of y $\frac{1}{2}$ mark for negative value of y and rejecting extraneous root**3 marks**

Question 14

T1

Angle θ , measuring $\frac{5\pi}{4}$, is drawn in standard position as shown below.

Determine the measures of all angles in the interval $[-4\pi, 2\pi]$ that are coterminal with θ .

**Solution**

$$\theta = -\frac{3\pi}{4} \quad \frac{1}{2} \text{ mark}$$

$$\theta = -\frac{11\pi}{4} \quad \frac{1}{2} \text{ mark}$$

1 mark

Prove the identity below for all permissible values of x :

$$\frac{\sin^2 x}{\sec x + 1} = \cos x - \cos^2 x$$

Solution

Method 1

$$\begin{aligned}\text{LHS} &= \frac{1 - \cos^2 x}{\frac{1}{\cos x} + 1} \\&= \frac{1 - \cos^2 x}{\frac{1 + \cos x}{\cos x}} \\&= (1 - \cos^2 x) \left(\frac{\cos x}{1 + \cos x} \right) \\&= (1 - \cos x)(1 + \cos x) \left(\frac{\cos x}{1 + \cos x} \right) \\&= (1 - \cos x)(\cos x) \\&= \cos x - \cos^2 x \\&= \text{RHS}\end{aligned}$$

1 mark for correct substitution of identities
1 mark for algebraic strategies
1 mark for logical process to prove the identity

3 marks

Method 2

$$\begin{aligned}\text{LHS} &= \frac{\sin^2 x}{\sec x + 1} \cdot \frac{(\sec x - 1)}{(\sec x - 1)} \\&= \frac{\sin^2 x (\sec x - 1)}{\sec^2 x - 1} \\&= \frac{\sin^2 x (\sec x - 1)}{\tan^2 x} \\&= \frac{\sin^2 x (\sec x - 1)}{\frac{\sin^2 x}{\cos^2 x}} \\&= \cos^2 x (\sec x - 1) \\&= \cos^2 x \left(\frac{1}{\cos x} - 1 \right) \\&= \cos x - \cos^2 x \\&= \text{RHS}\end{aligned}$$

1 mark for correct substitution of identities
1 mark for algebraic strategies
1 mark for logical process to prove the identity

3 marks

Solve algebraically:

$${}_nC_2 = 4n + 5$$

Solution

$${}_nC_2 = 4n + 5$$

$$\frac{n!}{(n-2)!2!} = 4n + 5$$

½ mark for factorial notation

$$\frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}2!} = 4n + 5$$

½ mark for factorial expansion

½ mark for simplification of factorial

$$n(n-1) = 2!(4n+5)$$

$$n^2 - n = 8n + 10$$

$$n^2 - 9n - 10 = 0$$

½ mark for simplification

$$(n-10)(n+1) = 0$$

$$n = 10 \quad \cancel{n = -1}$$

½ mark for both values of n

½ mark for rejecting extraneous root

3 marks

Question 17

P2

How many different arrangements are possible when arranging all of the letters of the word SEPTEMBER?

a) $9!$

b) $6!3!$

c) $\frac{9!}{3!}$

d) $\frac{6!}{3!}$

Question 18

T1

Which one of the following angles terminates in Quadrant III?

a) 3 radians

b) $\frac{7\pi}{5}$ radians

c) -210°

d) 500°

Question 19

P4

There are 13 terms in the expansion of $(3x - y)^{2n}$. Determine the value of n .

a) 6

b) 6.5

c) 7

d) 26

Question 20

T4

Which of the following is true about the periods of the three functions below?

$$f(\theta) = 2 \sin 3 \left(\theta - \frac{\pi}{2} \right)$$

$$g(\theta) = \sin 3\theta + 6$$

$$k(\theta) = 3 \sin \theta + 6$$

a) The graphs of $f(\theta)$ and $g(\theta)$ have the same period.

b) The graphs of $g(\theta)$ and $k(\theta)$ have the same period.

c) All of the graphs have the same period.

d) None of the graphs have the same period.

Question 21

T5

Which of the following represents the general solution to the equation $\tan \theta = -1$?

a) $\theta = \frac{\pi}{4} + 2k\pi, k \in \mathbb{I}$

b) $\theta = \frac{\pi}{4} + k\pi, k \in \mathbb{I}$

c) $\theta = \frac{3\pi}{4} + 2k\pi, k \in \mathbb{I}$

d) $\theta = \frac{3\pi}{4} + k\pi, k \in \mathbb{I}$

Question 22

R4

If $(3, -2)$ is a point on the graph of $y = f(x)$, what point must be on the graph of $y = 2f(x+1)$?

a) $(4, -1)$

b) $(4, -4)$

c) $(2, 1)$

d) $(2, -4)$

Question 23

R9

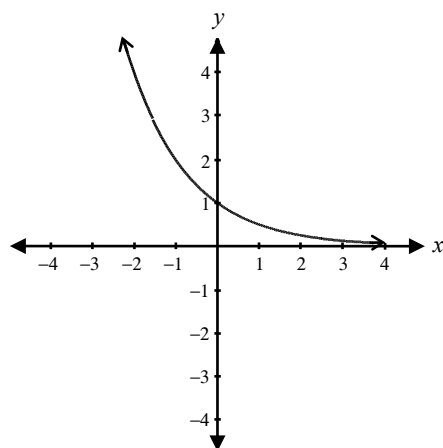
Which equation is represented by the graph sketched below?

a) $y = \left(\frac{1}{2}\right)^{-x}$

b) $y = \left(\frac{1}{2}\right)^x$

c) $y = 2^x$

d) $y = -2^x$



Question 24

R12

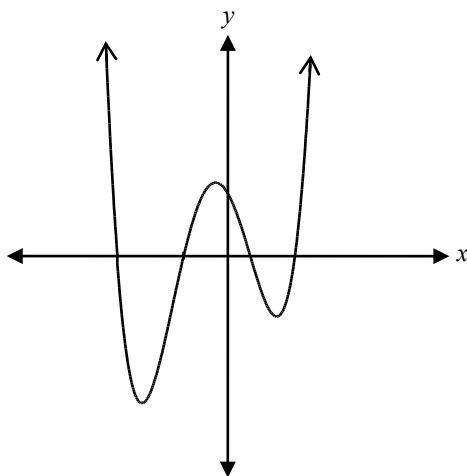
What is the degree of the polynomial represented below?

a) 2

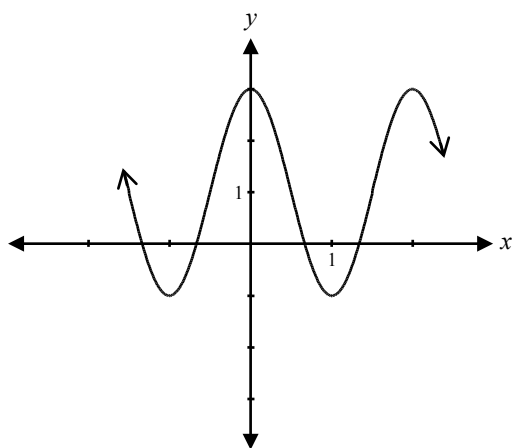
b) 3

c) 4

d) 5



Given the graph of $y = 2\cos \pi x + 1$ below, determine another equation that will produce the same graph.



Solution

Some sample equations are:

$$y = 2\cos \pi(x - 2) + 1$$

1 mark for correct equation

$$y = -2\cos \pi(x - 1) + 1$$

1 mark

$$y = -2\cos \pi(x + 1) + 1$$

$$y = 2\sin \pi\left(x + \frac{1}{2}\right) + 1$$

$$y = 2\sin \pi\left(x - \frac{3}{2}\right) + 1$$

Other answers are possible.

Question 26

R1

Given $f(x) = 3$ and $g(x) = x + 2$, determine the domain and range of $h(x) = \frac{f(x)}{g(x)}$.

Solution

Domain: $\{x \mid x \in \mathbb{R}, x \neq -2\}$

1 mark for domain

Range: $\{y \mid y \in \mathbb{R}, y \neq 0\}$

1 mark for range

2 marks

Question 27

T3

Explain how to find the exact value of $\sec\left(\frac{19\pi}{6}\right)$.

Solution

Find the exact value of $\cos\left(\frac{19\pi}{6}\right)$.

1 mark for $\cos\left(\frac{19\pi}{6}\right)$

Then take the reciprocal of the value of $\cos\left(\frac{19\pi}{6}\right)$.

1 mark for reciprocal

2 marks

Given $f(x) = 4 - x$, verify that $f^{-1}(x) = f(x)$.

Solution

Method 1

$$y = 4 - x$$

To find $f^{-1}(x)$, switch x and y values.

$$x = 4 - y$$

$$-y = x - 4$$

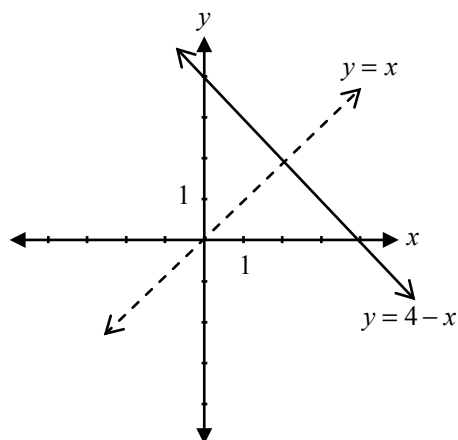
$$y = 4 - x$$

$$f^{-1}(x) = 4 - x$$

1 mark for verifying $f^{-1}(x) = f(x)$

1 mark

Method 2



When $y = 4 - x$ is reflected over the line $y = x$ it produces the same graph.

1 mark

Method 3

Assume $f^{-1}(x) = 4 - x$.

$$f(f^{-1}(x)) = 4 - (4 - x)$$

$$= x$$

$\therefore f(x)$ and $f^{-1}(x)$ are inverses of one another.

1 mark

Sketch the graph of:

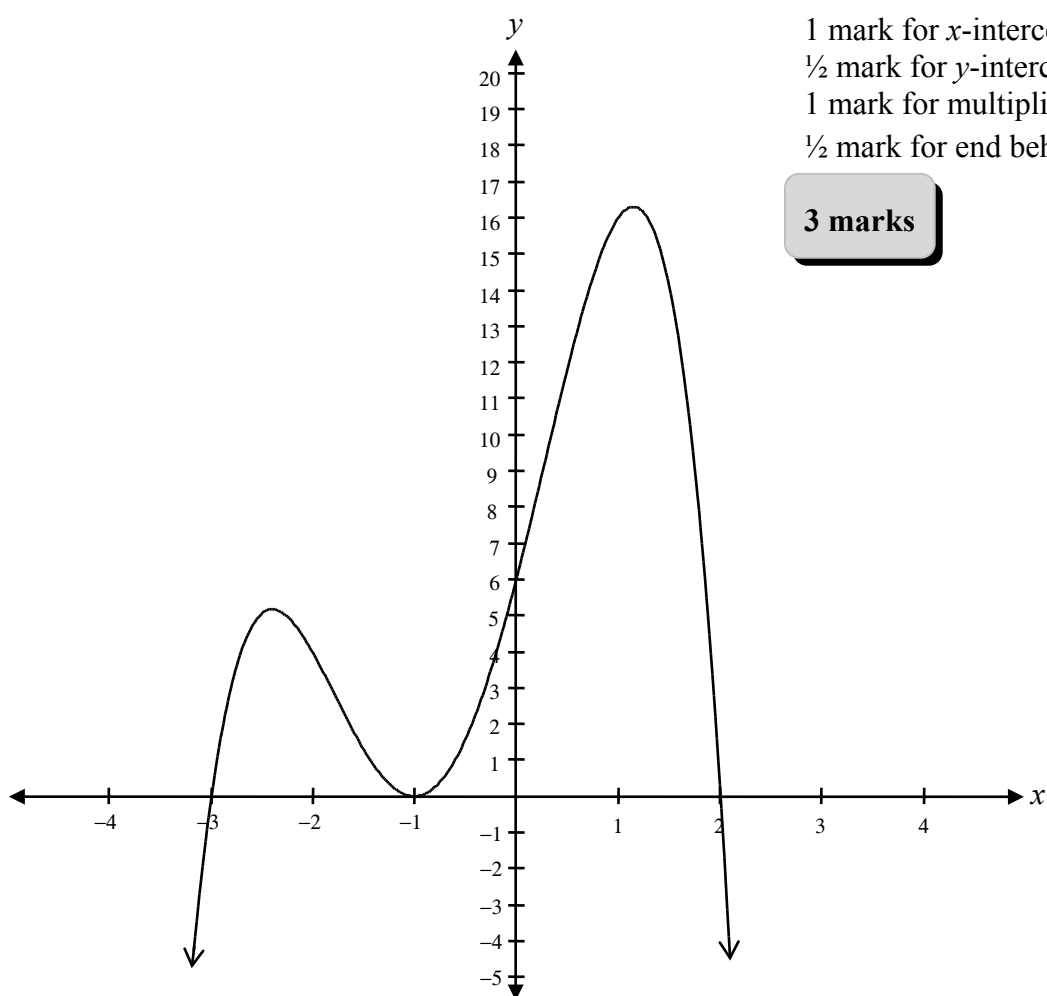
$$f(x) = (2 - x)(x + 3)(x + 1)^2$$

Label the x -intercepts and y -intercept.

Solution

x -intercepts: -3 , -1 , and 2

y -intercept: 6



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity of 2 at $x = -1$ only

$\frac{1}{2}$ mark for end behaviour

3 marks

Which expression has a larger value?

$$\log_2 36 \text{ or } \log_3 80$$

Justify your answer.

Solution

Method 1

$$\log_2 36 \quad 2^? = 36 \begin{cases} 2^5 = 32 \\ 2^6 = 64 \end{cases} \approx 5.1$$

$$\log_3 80 \quad 3^? = 80 \begin{cases} 3^3 = 27 \\ 3^4 = 81 \end{cases} \approx 3.9$$

$\therefore \log_2 36$ is the larger value

1 mark for justification

1 mark

Method 2

$\log_2 32 = 5 \therefore \log_2 36$ is a little more than 5

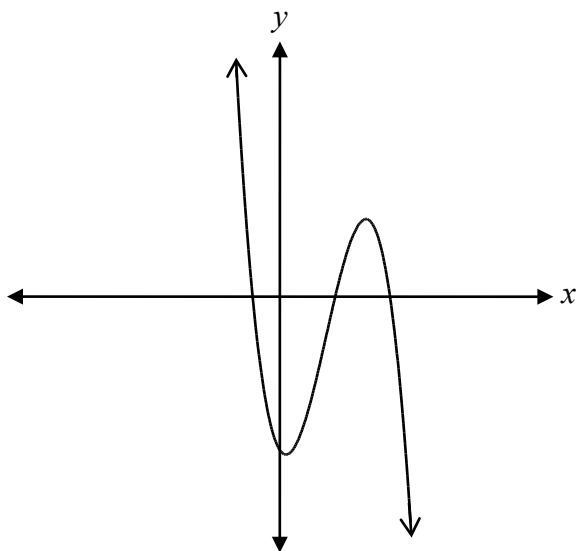
$\log_3 81 = 4 \therefore \log_3 80$ is a little less than 4

$\therefore \log_2 36$ is the larger value

1 mark for justification

1 mark

The graph below represents the equation $y = ax^3 + 6x^2 + 5x - 10$.



What must be true about the value of a ? Explain your reasoning.

Solution

a is any negative number.

$\frac{1}{2}$ mark

Explanation with reference to end behaviour.

$\frac{1}{2}$ mark for explanation

1 mark

or

a cannot be zero.

$\frac{1}{2}$ mark

The graph is of a cubic function,
not a quadratic function.

$\frac{1}{2}$ mark for explanation

1 mark

The terminal arm of an angle θ , in standard position, intersects the unit circle in Quadrant IV at a point

$P\left(\frac{\sqrt{5}}{4}, y\right)$. Determine the value of $\sin \theta$.

Solution

Method 1

The point $P(\theta)$ on the unit circle has coordinates $(\cos \theta, \sin \theta)$.

$$\cos^2 \theta + \sin^2 \theta = 1$$

$\frac{1}{2}$ mark for showing $y = \sin \theta$

$$\left(\frac{\sqrt{5}}{4}\right)^2 + \sin^2 \theta = 1$$

$\frac{1}{2}$ mark for substitution

$$\sin^2 \theta = 1 - \frac{5}{16}$$

$$\sqrt{\sin^2 \theta} = \sqrt{\frac{11}{16}}$$

$$\sin \theta = \pm \frac{\sqrt{11}}{4}$$

$\frac{1}{2}$ mark for solving for $\sin \theta$

$$\sin \theta = -\frac{\sqrt{11}}{4}$$

$\frac{1}{2}$ mark for a negative $\sin \theta$ value in Quadrant IV

2 marks

Method 2

$$(\sqrt{5})^2 + y^2 = 4^2$$

$\frac{1}{2}$ mark for substitution

$$5 + y^2 = 16$$

$$y^2 = 11$$

$$y = \pm \sqrt{11}$$

$\frac{1}{2}$ mark for solving for y

$$\sin \theta = -\frac{\sqrt{11}}{4}$$

$\frac{1}{2}$ mark for using the value of y to find the value of $\sin \theta$

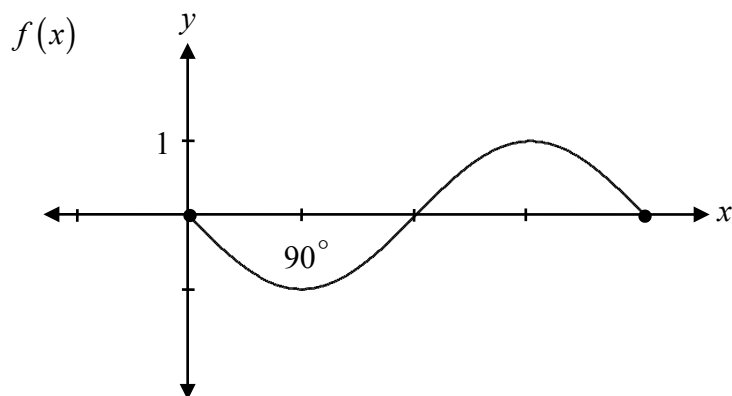
$\frac{1}{2}$ mark for a negative $\sin \theta$ value in Quadrant IV

2 marks

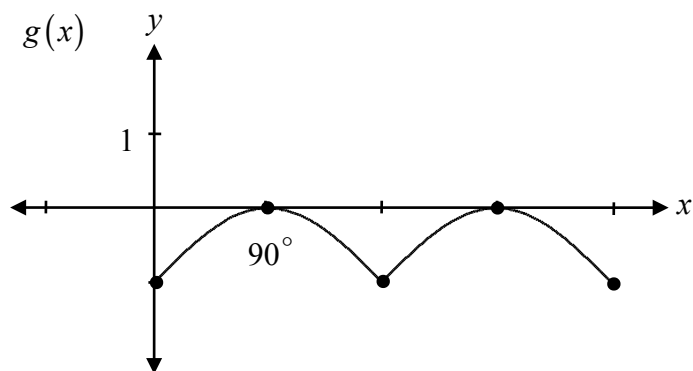
Question 33

R1, R2

Given the sinusoidal function $f(x)$ below, sketch the graph of $g(x) = |f(x)| - 1$.



Solution



1 mark for absolute value
1 mark for vertical shift

2 marks

Question 34

R14

The graph of a rational function, $f(x)$, has a point of discontinuity when $x = 2$ and an asymptote when $x = 4$. Write a possible equation for $f(x)$.

Solution

A possible equation is:

$$f(x) = \frac{x-2}{(x-2)(x-4)}$$

1 mark for $\frac{x-2}{x-2}$ (point of discontinuity when $x = 2$)

1 mark for $x-4$ in denominator (asymptote when $x = 4$)

2 marks

Question 35

R11

Given that $(x-1)$ is one of the factors, express $x^3 - 57x + 56$ as a product of factors.

Solution

$$\begin{array}{r|rrrr} 1 & 1 & 0 & -57 & 56 \\ & \downarrow & 1 & 1 & -56 \\ \hline & 1 & 1 & -56 & 0 \end{array}$$

$\frac{1}{2}$ mark for $x = 1$

1 mark for synthetic division (or for any other equivalent strategy)

$$(x-1)(x^2 + x - 56)$$

$\frac{1}{2}$ mark for consistent factors

or

$$(x-1)(x+8)(x-7)$$

2 marks

Give an example using values for A and B , in degrees or radians, to verify that $\cos(A + B) = \cos A + \cos B$ is **not** an identity.

Solution

Method 1

Let $A = 45^\circ$ and $B = 90^\circ$.

LHS	RHS
$\cos(45^\circ + 90^\circ)$	$\cos 45^\circ + \cos 90^\circ$
$\cos(135^\circ)$	$\cos 45^\circ + \cos 90^\circ$
$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2} + 0$
$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$

1 mark for simplification of $\cos(A + B)$

1 mark for simplification of $\cos A + \cos B$

2 marks

$\text{LHS} \neq \text{RHS} \therefore \cos(A + B) = \cos A + \cos B$ is not an identity.

Method 2

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

Let $A = 60^\circ$ and $B = 30^\circ$.

$$\cos(60^\circ + 30^\circ) = \cos 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ$$

$$= \left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right)$$

$$= \frac{\sqrt{3} - \sqrt{3}}{4}$$

$$= 0$$

1 mark for simplification of $\cos(A + B)$

$$\cos A + \cos B = \cos 60^\circ + \cos 30^\circ$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2}$$

$$= \frac{1 + \sqrt{3}}{2}$$

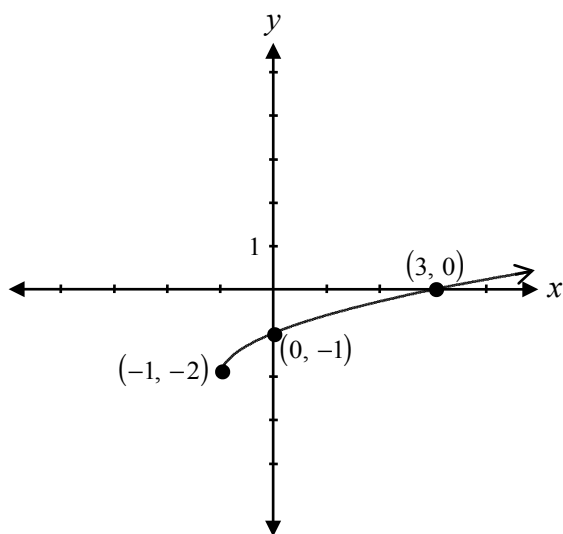
1 mark for simplification of $\cos A + \cos B$

2 marks

These two solutions are not equal $\therefore \cos(A + B) = \cos A + \cos B$ is not an identity.

Sketch the graph of $y = \sqrt{x+1} - 2$ and verify that the value of the x -intercept is the same as the solution to the equation $\sqrt{x+1} - 2 = 0$.

Solution



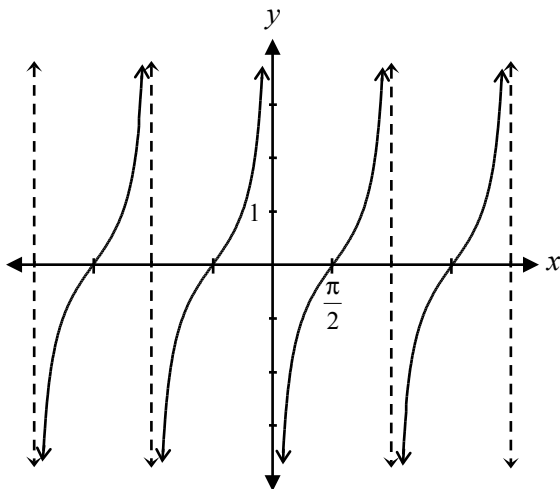
1 mark for general shape
 $\frac{1}{2}$ mark for horizontal shift
 $\frac{1}{2}$ mark for vertical shift

$$\begin{array}{ll}
 \sqrt{x+1} = 2 & \sqrt{x+1} - 2 = 0 \\
 (\sqrt{x+1})^2 = (2)^2 & \text{or} \quad \sqrt{3+1} - 2 = 0 \\
 x+1 = 4 & \sqrt{4} - 2 = 0 \\
 x = 3 & 0 = 0
 \end{array}$$

1 mark for verification

3 marks

Mohamed is asked to sketch the graph of $y = \tan x$.
His graph is shown below.



Explain why his graph is incorrect.

Solution

The graph of $y = \tan x$ should have zeros at $k\pi$, $k \in \mathbb{I}$.

1 mark for explanation

or

1 mark

The graph of $y = \tan x$ should have asymptotes at $(2k + 1)\frac{\pi}{2}$ or $\frac{\pi}{2} + k\pi$, $k \in \mathbb{I}$.

or

Mohamed sketched the incorrect graph. He sketched the graph of $y = \tan\left(x - \frac{\pi}{2}\right)$.

On the interval $0 \leq \theta < 2\pi$, identify the non-permissible values of θ for the trigonometric identity:

$$\tan \theta = \frac{1}{\cot \theta}$$

Solution

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{\frac{\cos \theta}{\sin \theta}}$$

$\therefore$ the above identity is non-permissible when $\cos \theta = 0$ or $\sin \theta = 0$.

1 mark for identifying non-permissible values
($\frac{1}{2}$ mark for $\cos \theta = 0$, $\frac{1}{2}$ mark for $\sin \theta = 0$)

$$\cos \theta \neq 0$$

$$\theta \neq \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\sin \theta \neq 0$$

$$\theta \neq 0, \pi$$

1 mark for solving for θ ($\frac{1}{2}$ mark for each solution set)

2 marks

$$\theta \neq 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$$

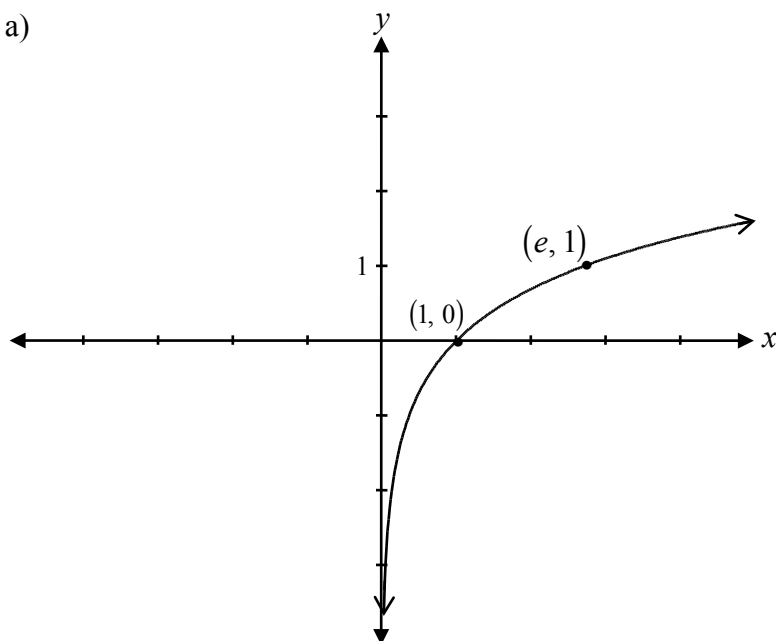
Question 40

a) R9 b) R2, R5

- a) Sketch the graph of $y = \ln(x)$.
 b) Sketch the graph of $y = -\ln(x - 2)$.

Solutions

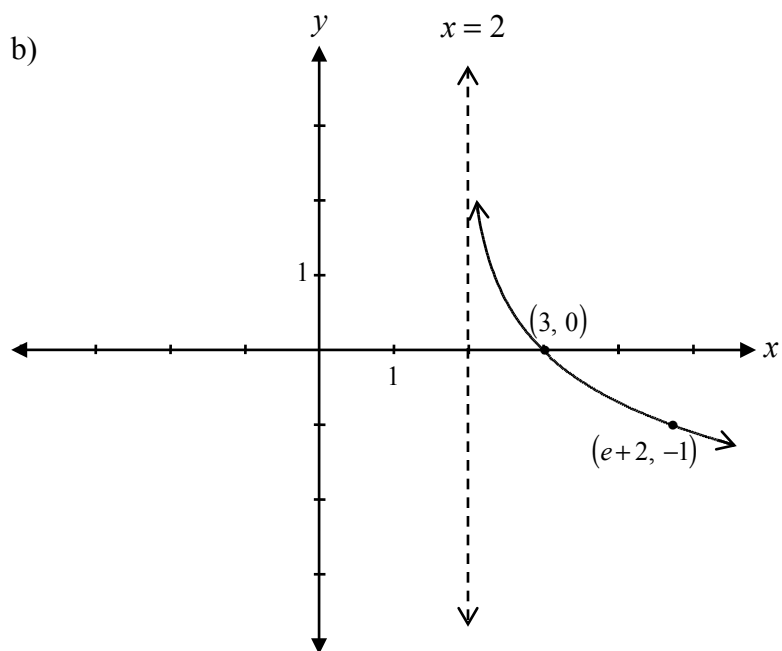
a)



$\frac{1}{2}$ mark for increasing logarithmic function
 $\frac{1}{2}$ mark for x -intercept at $(1, 0)$
 $\frac{1}{2}$ mark for consistent point on logarithmic function
 $\frac{1}{2}$ mark for vertical asymptotic behaviour

2 marks

b)



1 mark for reflection in x -axis
 1 mark for horizontal shift

2 marks

Given $f(x) = \sqrt{x-2}$ and $g(x) = 3x$, write the equation for $h(x) = f(g(x))$.

What are the restrictions on the domain of $h(x)$?

Explain your reasoning.

Solution

$$h(x) = \sqrt{3x-2}$$

1 mark for $h(x) = f(g(x))$

$$3x - 2 \geq 0$$

$$3x \geq 2$$

$$x \geq \frac{2}{3}$$

$\frac{1}{2}$ mark for identifying restriction

Since we cannot find a square root of a negative number, there is a restriction

$\frac{1}{2}$ mark for explanation

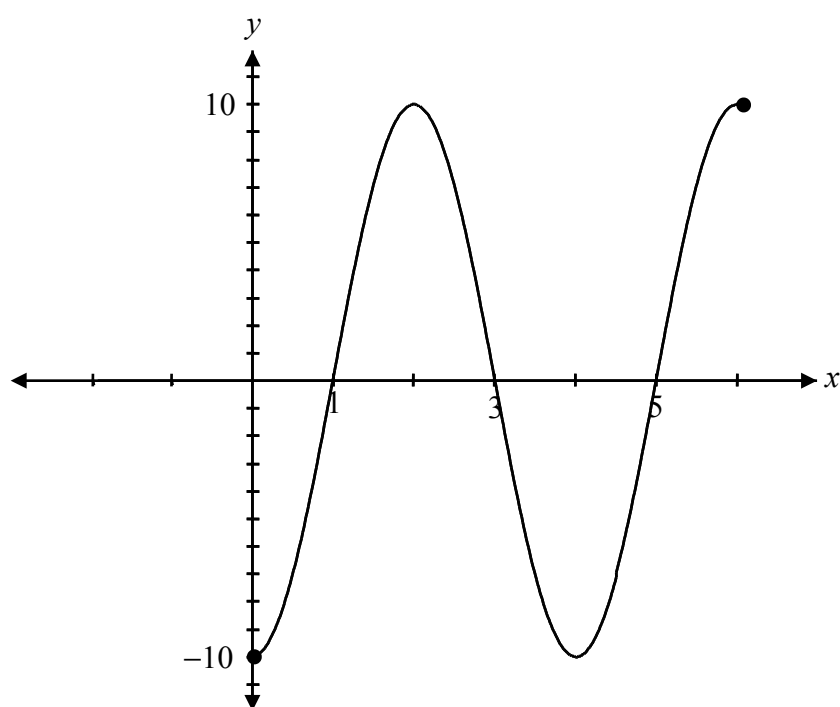
on the domain, $x \geq \frac{2}{3}$.

2 marks

Sketch the graph of $y = 10 \cos\left[\frac{\pi}{2}(x - 2)\right]$ over the interval $[0, 6]$.

Solution

$$\text{period} = \frac{2\pi}{\frac{\pi}{2}} = 4$$



1 mark for amplitude
1 mark for period
1 mark for horizontal shift

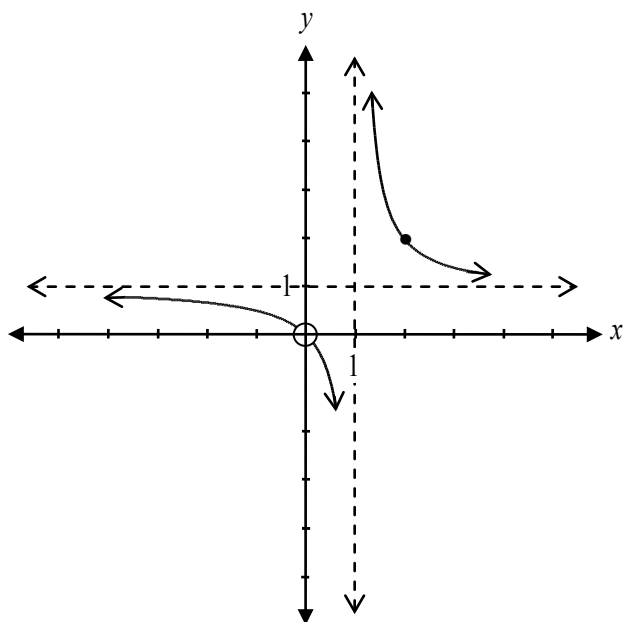
3 marks

Note(s):

- deduct ½ mark if the interval $[0, 6]$ is not completely sketched

Sketch the graph of the function $f(x) = \frac{x^2}{x^2 - x}$.

Solution



$$f(x) = \frac{x^2}{x(x-1)}$$

$$= \frac{x}{x-1} \text{ with a point of discontinuity where } x = 0$$

$$\text{point of discontinuity: } f(0) = \frac{0}{0-1} = 0$$

$\therefore$ there is a point of discontinuity at $(0, 0)$.

divide:

$$\begin{array}{r} 1 \\ x-1 \overline{) x+0} \\ \underline{x-1} \\ 1 \end{array}$$

or

$$\begin{aligned} f(x) &= \frac{x}{x-1} \\ &= \frac{x-1+1}{x-1} \\ &= 1 + \frac{1}{x-1} \end{aligned}$$

$$\therefore f(x) = \frac{1}{x-1} + 1$$

$\therefore$ horizontal asymptote at $y = 1$

$\therefore$ vertical asymptote at $x = 1$

1 mark for vertical asymptote at $x = 1$

1 mark for horizontal asymptote at $y = 1$

1 mark for point of discontinuity at $(0, 0)$ or a point of discontinuity consistent with graph

$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

4 marks

Is $(x - 3)$ a factor of $x^4 - x^3 - 3x^2 + x - 1$?

Justify your answer.

Solution

Method 1

$$x = 3$$

$$\therefore (3)^4 - (3)^3 - 3(3)^2 + (3) - 1 = 81 - 27 - 27 + 3 - 1 = 29$$

The remainder does not equal zero,
therefore $(x - 3)$ is not a factor.

$\frac{1}{2}$ mark for $x = 3$

1 mark for remainder theorem

$\frac{1}{2}$ mark for explanation

2 marks

Method 2

3	1	-1	-3	1	-1
↓	3	6	9	30	
1	2	3	10	29	

The remainder does not equal zero,
therefore $(x - 3)$ is not a factor.

$\frac{1}{2}$ mark for $x = 3$

1 mark for synthetic division

$\frac{1}{2}$ mark for explanation

2 marks

Given $f(x) = x - 1$ and $g(x) = x^2$, write the equation of $y = f(g(x))$ and sketch the graph.

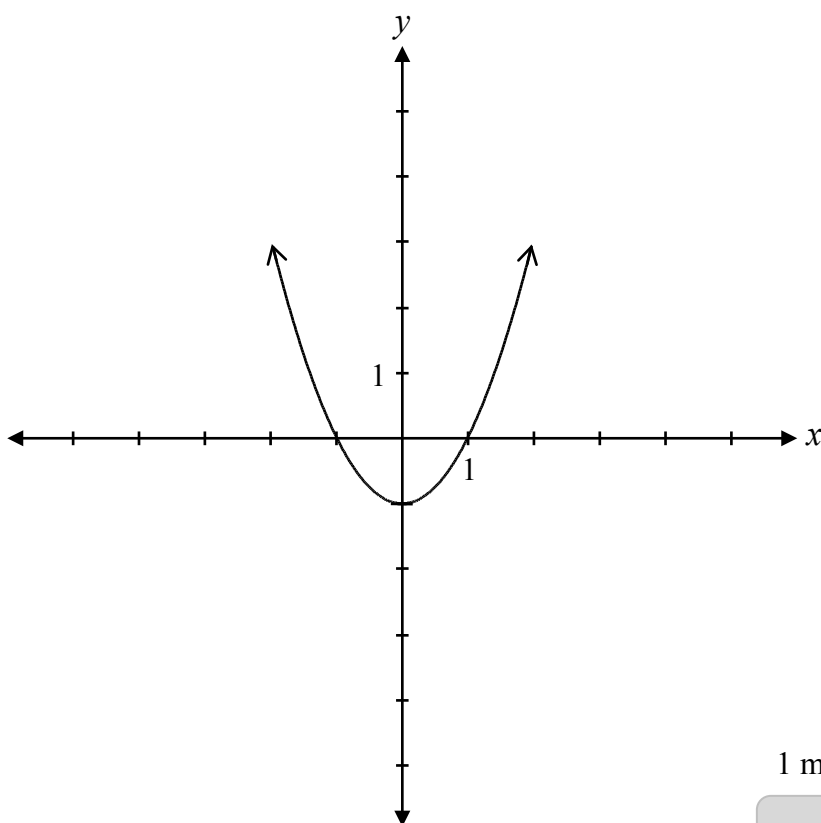
Solution

$$f(g(x)) = x^2 - 1$$

1 mark for composition

or

$$y = x^2 - 1$$



1 mark for consistent graph

2 marks